

# Effect of Notch1 signaling on muscle engraftment and maturation from pluripotent stem cells

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## SUMMARY

Pax3-induced pluripotent stem cell-derived myogenic progenitors display an embryonic molecular signature but become postnatal upon transplantation. Because this correlates with upregulation of Notch signaling, here we probed whether NOTCH1 is required for *in vivo* maturation by performing gain- and loss-of-function studies in inducible Pax3 (iPax3) myogenic progenitors. Transplantation studies revealed that Notch1 signaling did not change the number of donor-derived fibers; however, the NOTCH1 overexpression cohorts showed enhanced satellite cell engraftment and more mature fibers, as indicated by fewer fibers expressing the embryonic myosin heavy-chain isoform and more type IIX fibers. While donor-derived Pax7+ cells were detected in all transplants, in the absence of Notch1, secondary grafts exhibited a high fraction of these cells in the interstitial space, indicating that NOTCH1 is required for proper satellite cell homing. Transcriptional profiling of NOTCH1-modified donor-derived satellite cells suggests that this may be due to changes in the extracellular matrix organization, cell cycle, and metabolism.

## INTRODUCTION

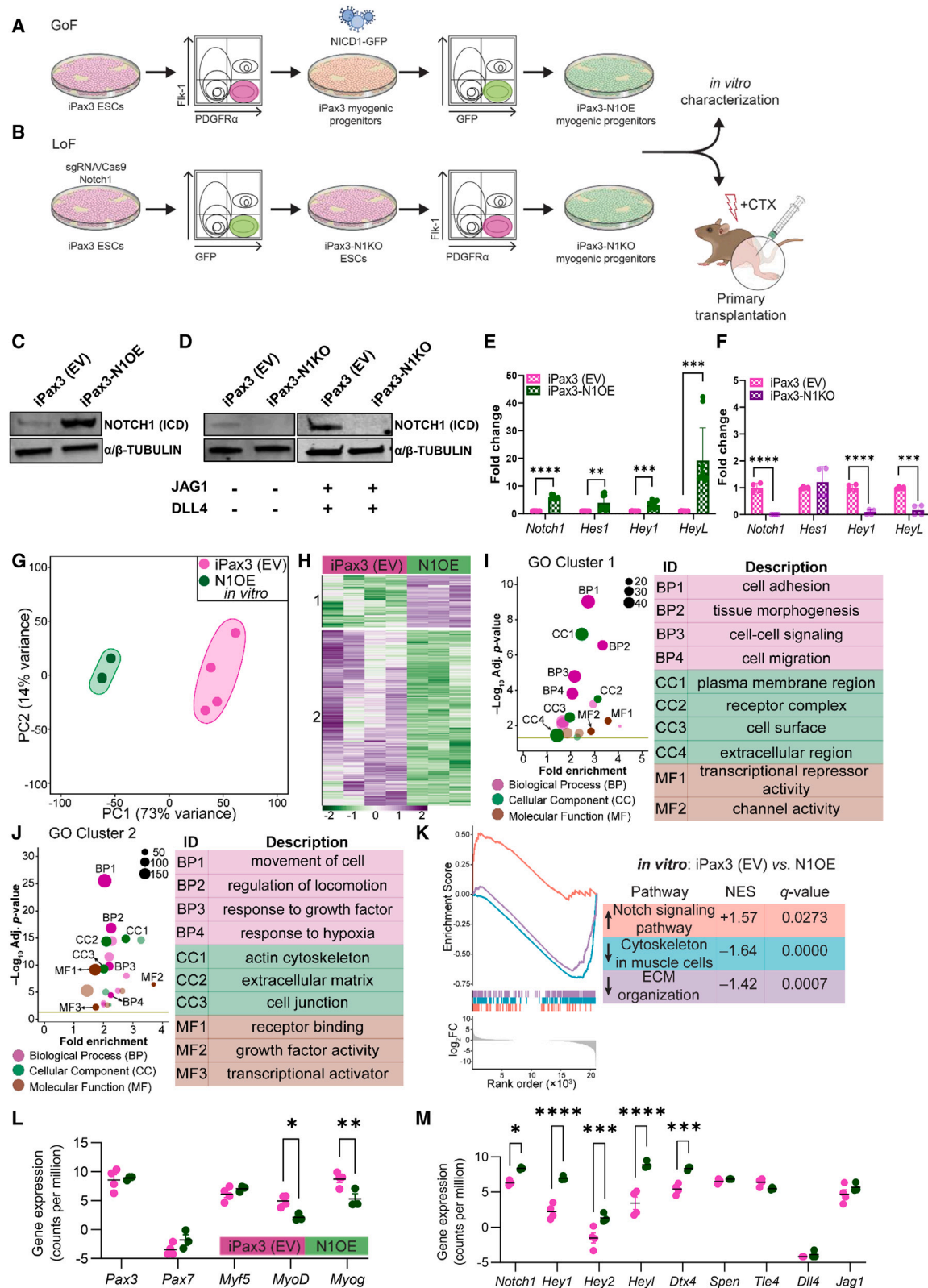
Pluripotent stem cells (PSCs) possess unique features, such as unlimited proliferative potential and ability to differentiate into all cell types, thus representing an attractive system for generating cells for therapeutic application. We have shown that conditional expression of PAX3 or PAX7 during the appropriate development window of *in vitro* PSC differentiation enables the symmetrical expansion of large numbers of inducible Pax3 (iPax3)/iPax7 myogenic progenitors, which upon transplantation give rise to muscle fibers and also stem cells (known as satellite cells), thus enabling sustained engraftment (Azzag et al., 2020, 2022; Darabi et al., 2008, 2011, 2012; Filareto et al., 2012, 2013). Transcriptomic analysis of these *in-vitro*-generated myogenic progenitors prior to transplantation revealed a prenatal molecular profile, similar to embryonic/fetal myoblasts (Incitti et al., 2019). However, this changed upon exposure to the adult muscle environment, as engrafted donor-derived CD34<sup>+</sup> putative satellite cells exhibited a transcriptome resembling postnatal satellite cells (Incitti et al., 2019). Among many signaling pathway changes, we observed upregulation of several components of Notch signaling in re-isolated donor-derived CD34<sup>+</sup> cells, suggesting a potential role for Notch signaling in the *in vivo* maturation switch of PSC-derived myogenic cells.

The Notch signaling pathway is an evolutionarily conserved intracellular signaling system that plays critical roles in tissue morphogenesis during the development

and homeostasis of multiple tissues and organs postnatally (Hori et al., 2013), including the skeletal muscle (Mourikis and Tajbakhsh, 2014). In vertebrates, Notch signaling is initiated upon interaction of its extracellular ligands (Dll1 and -4 and Jag1 and -2) with Notch receptors (Notch1, -2, -3, and -4), which leads to cleavage of Notch intracellular domain (NICD) (Artavanis-Tsakonas et al., 1999). The NICD translocate to the nucleus, interacts with RBP-J, and activates transcription target genes, including Hes and Hey family genes (Vasyutina et al., 2007). In the absence of NICD, RBP-J binds to transcriptional repressors, inhibiting transcription of target genes (Jarriault et al., 1995). In the adult, Notch1 signaling is mostly known for its function in the maintenance and self-renewal of satellite cells (Brack et al., 2008; Pasut et al., 2016; Wen et al., 2012). Despite extensive research on the role of Notch1 signaling in satellite cell quiescence and muscle regeneration, its involvement in muscle maturation from PSCs has not been investigated.

Here, we performed gain-of-function (GoF) and loss-of-function (LoF) studies targeting the Notch1 receptor in PSC-derived myogenic progenitors to probe whether Notch1 signaling is required for their *in vivo* maturation switch. Our findings show that Notch1 GoF in iPax3 myogenic progenitors enhances satellite cell engraftment and produces more mature muscle fibers. On the other hand, lack of NOTCH1 resulted in myofiber and satellite cell engraftment comparable to iPax3 controls, but displayed lower numbers of oxidative fibers. Of note,





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secondary transplantation assays revealed a striking phenotype for the NOTCH1 LoF, as a high frequency of donor-derived Pax7+ cells were found at the interstitial space, suggesting that NOTCH1 is required for proper satellite cell seeding. Transcriptional profiling of iPax3 donor-derived satellite cells with altered Notch signaling revealed significant changes in several pathways, in particular genes related to the extracellular matrix (ECM) organization, cell cycle, and metabolism.

## RESULTS

### NOTCH1 GoF and LoF in iPax3 myogenic progenitors

To determine whether Notch1 is important for the *in vivo* maturation of PSC-derived iPax3 myogenic progenitors, here we investigated the effect of NOTCH1 GoF and LoF in iPax3 myogenic progenitors by overexpressing NICD1 (iPax3-N1OE) or ablation of *Notch1* (iPax3-N1KO), respectively. For the generation of iPax3-N1OE myogenic progenitors, we transduced iPax3 myogenic progenitors with lentiviral vectors constitutively expressing NICD1 tagged with GFP. An empty vector (EV)-tagged with GFP served as control. Modified cells were subsequently purified by fluorescence-activated cell sorting (FACS) based on GFP expression (Figures 1A and S1A). For iPax3-N1KO, we modified PSCs using CRISPR-Cas9 in combination with a vector expressing a guide RNA targeting *Notch1* or a control GFP-tagged EV (Figure 1B). Transduced cells were purified based on GFP expression (Figure S1B) and subsequently differentiated into myogenic progenitors (Figure 1B). The absence of Notch1 did not affect the ability of iPax3 PSC to give rise to myogenic progenitors (Figures S1C and S1D).

Using western blot, we confirmed the increased protein expression of NOTCH1 in iPax3-N1OE myogenic progenitors (Figure 1C) compared to control iPax3-EV counterparts. For the LoF, NICD1 was not detected in cultures of

both iPax3-N1KO and iPax3-EV cells (Figure 1D, left panel). However, when we induced Notch signaling by treatment with the NOTCH ligands DLL4 and JAG1, NICD1 became evident in iPax3-EV control but not in iPax3-N1KO cultures (Figure 1D, right panel), thus confirming the elimination of NOTCH1. Consistently, gene expression analyses showed significant upregulation of *Notch1* as well as its downstream target genes *Hes1*, *Hey1*, and *HeyL* in the iPax3-N1OE cohort (Figure 1E), whereas the iPax3-N1KO displayed the opposite phenotype with blunted levels of *Notch1* and respective downstream target genes *Hey1* and *HeyL* (Figure 1F). For subsequent comparison assessments, iPax3 (EV) control results from both arms were pulled together.

Having confirmed the GoF and LoF phenotypes of NOTCH1-modified iPax3 myogenic progenitors, we subjected these cells to *in vitro* terminal differentiation. All samples gave rise to myosin heavy-chain (MHC)-expressing myotubes (Figure S1E), but differentiation was impaired in the iPax3-N1OE cohort, demonstrated by the quantification of fusion index, myotube width, and the percentage of myotube area (Figures S1F–S1H). High levels of GFP expression, measured by flow cytometry and immunofluorescence, confirmed the maintenance of NICD1 expression in N1OE myogenic progenitors (Figures S1I–S1K) and myotubes (Figure S1L). Quantification of MYOD+ and Myogenin+ (MYOG) nuclei in myogenic progenitor and myotube cultures showed overall lower expression of MYOD and MYOG in N1OE when compared to controls (Figures S2A–S2D). Co-staining with GFP confirmed that all cells were double-positive (MYOD+GFP+ and MYOG+GFP+), both with and without N1OE (Figures S2E–S2H). These *in vitro* data showing that Notch1 overexpression leads to impaired fusion and downregulation of MYOD and MYOG align with a previous study by Sakai and colleagues (Sakai et al., 2017), who reported that ligand-induced *in vitro* activation of Notch signaling in mouse myogenic cells and human myoblasts results in

### Figure 1. Characterization of iPax3-N1OE and iPax3-N1KO myogenic progenitors

(A and B) Schemes outline GoF (A) and LoF (B) studies in PSC-derived iPax3 myogenic progenitors.

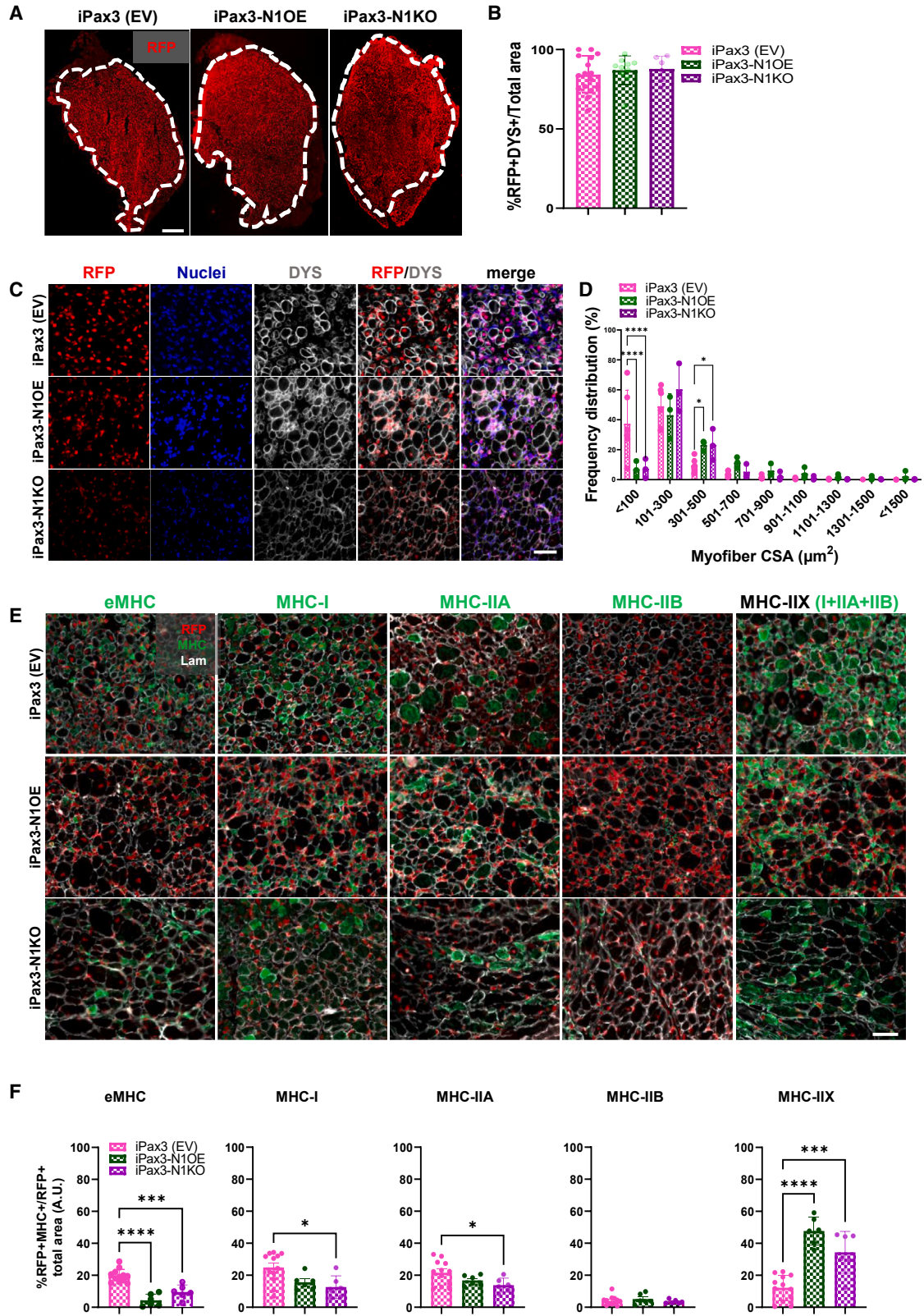
(C and D) Western blot for the intracellular domain (ICD) of NOTCH1 (120 kD), in iPax3-N1OE (C) and iPax3-N1KO (D) myogenic progenitors along with their respective iPax3 control counterparts.  $\alpha$ / $\beta$ -TUBULIN served as loading control (55 kD). These are representative images of three (iPax3-N1OE) and two (iPax3-N1KO) independent experiments.

(E and F) Bar graphs show gene expression for *Notch1* and downstream target genes: *Hes1*, *Hey1*, and *HeyL* in iPax3-N1OE (E) and iPax3-N1KO (F) myogenic progenitors along with iPax3 controls. Results are normalized to *Gapdh*. Data are shown as mean  $\pm$  SEM of 3 independent experiments. \*\* $p$  < 0.01, \*\*\* $p$  < 0.001, and \*\*\*\* $p$  < 0.0001 by unpaired Student's *t* test.

(G–K) RNA sequencing analysis. (G) Principal component analysis for iPax3-N1OE (green) and iPax3 control (pink) shows the distribution of samples according to DEGs. (H) Heatmap representing hierarchical clustering of the top 1,454 DEGs shared among all samples. (I) Cluster 1 GO analysis. (J) Cluster 2 GO analysis. (K) GSE analysis with upregulated and downregulated KEGG pathways between iPax3 (EV) and iPax3-N1OE myogenic progenitors.

(L and M) Bar graphs show gene expression for myogenic markers (L) and components of Notch1 signaling (M) in iPax3-N1OE and iPax3 controls. Data are shown as mean  $\pm$  SEM of 4 and 3 independent experiments for iPax3 controls (magenta) and N1OE (green), respectively. \* $p$  < 0.05, \*\* $p$  < 0.01, \*\*\* $p$  < 0.001, and \*\*\*\* $p$  < 0.0001 by ANOVA with Dunnett's correction.





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the repression of MYOD and MYOG. In contrast, N1KO showed higher numbers of MYOD+ and MYOG+ cells at the progenitor stage (Figures S2A and S2B) and reduced expression in differentiated myotubes (Figures S2C and S2D). These results are in agreement with the notion that conditional ablation of *Notch1* in satellite cells leads to precocious differentiation (Brohl et al., 2012; Fujimaki et al., 2018).

### Transcriptomics of iPax3-N1OE myogenic progenitors

To determine the molecular changes underlying NOTCH1 GoF in iPax3 myogenic progenitors, we performed RNA sequencing. Linear dimensionality reduction analysis demonstrated clear separation along the first principal component (PC1) axis between iPax3-N1OE and iPax3 controls (Figure 1G). Unsupervised hierarchical clustering analysis of the top 1,454 differentially expressed genes (DEGs) revealed two distinct clusters (Figure 1H). Gene ontology (GO) of these clusters showed significant enrichment of biological process related to cell adhesion, tissue morphogenesis, cell-cell signaling, and cell migration in cluster 1 (Figure 1I) and movement of cell, regulation of locomotion, response to growth factor, and response to hypoxia in cluster 2 (Figure 1J). When we compared iPax3-N1OE to iPax3 control by gene set enrichment analysis (GSEA) using Kyoto Encyclopedia of Genes and Genomes (KEGG)-annotated signaling pathways, we found upregulation of genes associated with Notch signaling pathway and downregulation of genes associated with ECM organization and cytoskeleton in muscle cells (Figure 1K). In agreement with the immunofluorescence results described previously (Figures S2A and S2B), and a previous report of NICD<sup>OE</sup> in satellite cells (Wen et al., 2012), iPax3 myogenic progenitors overexpressing NICD1 show lower levels of *MyoD* and *Myogenin* (Figure 1L), correlating with their impaired ability to differentiate into myotubes *in vitro* (Figures S1E–S1H). Pax3-induced myogenic progenitors do not express *Pax7* *in vitro* (Filareto et al., 2013), and N1OE did not show a significant effect on *Pax7* expression levels (Figure 1L). Likewise, no differences were observed in *Pax3* expression, which is expected since both cohorts have

*Pax3* expression induced by doxycycline (Figure 1L). Regarding Notch signaling, the RNA sequencing results confirm the upregulation of *Notch1* and Notch target genes *Hey1* and *HeyL*, in addition to *Hey2* and the E3 ubiquitin ligase *Dtx4* (Figure 1M).

### Engraftment and fiber type composition upon NOTCH1 modification

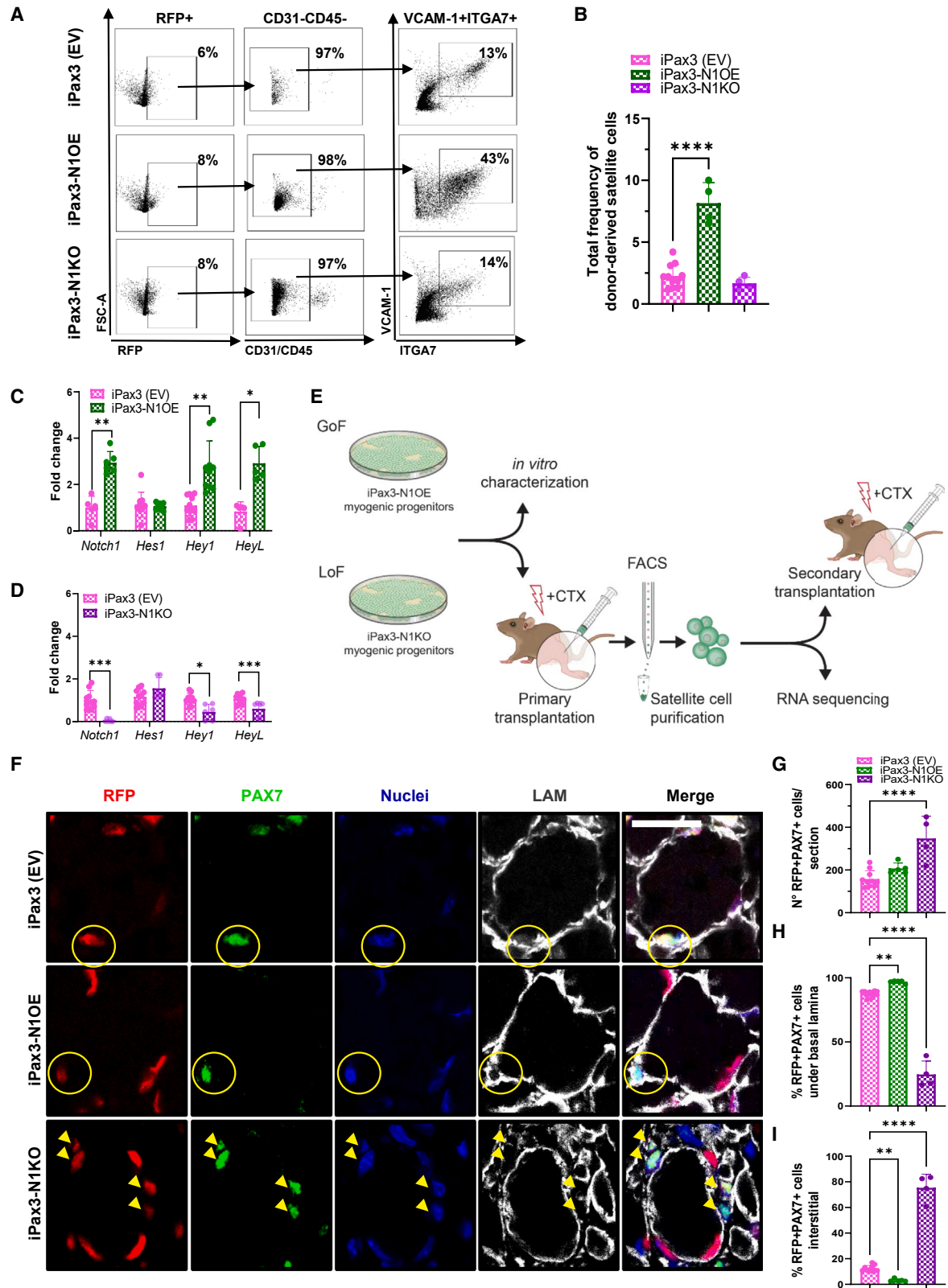
To determine the effect of NOTCH1 GoF and LoF on the *in vivo* muscle regenerative potential of PSC-derived myogenic progenitors, we transplanted 300,000 iPax3-N1OE, iPax3-N1KO, and respective iPax3 control myogenic progenitors into irradiated and cardiotoxin (CTX)-injured muscles of NSG mice (as outlined in Figure 1A). Hindlimbs received a 12-Gy single dose, and 24 h later, both tibialis anterior (TA) muscles were injured with 10  $\mu$ M CTX. Transplantation was performed next day. Four weeks post transplantation, we evaluated engraftment through immunostaining with RFP (donor-tracking) and dystrophin antibodies. Our results showed comparable myofiber engraftment between iPax3-EV iPax3-N1OE and iPax3-N1KO cohorts (Figures 2A and 2C), which was confirmed by the quantification of donor-derived RFP+DYS+ myofibers (Figure 2B). Similar engraftment was observed upon the transplantation of non-transduced control iPax3 myogenic progenitors (Figures S3A–S3C). Quantification of myofiber cross-sectional area (CSA) revealed that both N1OE and N1KO cohorts display reduced frequency of small myofibers (<100  $\mu$ m<sup>2</sup>) and increased frequency of myofibers in the range of 300–500  $\mu$ m<sup>2</sup> when compared to controls (Figure 2D).

To determine whether NOTCH1 GoF or LoF affects myofiber graft composition, we performed immunostaining for different MHC isoforms, including embryonic (eMHC), oxidative slow-twitch type I (MHC-I), oxidative fast-twitch type IIA (MHC-IIA), as well as glycolytic fast-twitch type IIB (MHC-IIB), and by exclusion, glycolytic fast-twitch type IIX (MHC-IIX). iPax3-N1OE muscle grafts displayed significantly lower expression of eMHC, while the frequency of IIX fibers was increased (Figures 2E and 2F), suggesting that NOTCH1 overexpression enhances *in vivo* maturation.

### Figure 2. Myofiber engraftment upon the transplantation of iPax3-N1OE and iPax3-N1KO myogenic progenitors

(A–D) Representative images show myofiber engraftment in muscles that had been transplanted with iPax3, iPax3-N1OE, and iPax3-N1KO myogenic progenitors. Scale bar, 500  $\mu$ m. Staining for RFP in red and dystrophin (DYS) in white. DAPI-stained nuclei (A, C). Graph shows the area of RFP+DYS+ myofiber engraftment per muscle total area (B). Data are presented as mean  $\pm$  SEM of two independent transplantation experiments ( $n$  = 16 TA muscles for iPax3 controls,  $n$  = 10 TAs for N1OE, and  $n$  = 6 TAs for N1KO). (D) Graph shows the frequency distribution of myofiber CSA area ( $n$  = 6 TAs for iPax3 controls,  $n$  = 3 TAs for N1OE, and  $n$  = 3 TAs for N1KO). \* $p$  < 0.05 and \*\*\*\* $p$  < 0.0001 by ANOVA with Dunnett's correction.

(E and F) Representative images show staining for different MHC isoforms (AF647 signal, rendered in green), RFP (in red), and laminin (LAM, in white). Scale bar, 100  $\mu$ m (F). Bar graphs show respective quantification (E), as the ratio between RFP+ engraftment area and MHC expression ( $n$  = 12 TAs for iPax3 controls,  $n$  = 6 TAs for N1OE, and  $n$  = 6 TAs for N1KO). \* $p$  < 0.05, \*\*\* $p$  < 0.001, and \*\*\*\* $p$  < 0.0001 by ANOVA with Dunnett's correction.



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Changes in myofiber type were also observed in the iPax3-N1KO cohort, which displayed lower expression of eMHC, MHC-I, and MHC-IIA, while IIX fibers were increased (Figures 2E and 2F), suggesting that Notch1 signaling may be of importance for the development/maintenance of oxidative fibers. Based on the unexpected results for eMHC (similar trend to iPax3-N1OE), we investigated whether NOTCH1 LoF resulted in upregulation of other Notch receptors, such as Notch2 and Notch3. Our data suggest that this is not the case, as expression levels for these receptors were comparable to iPax3 controls (Figures S3D and S3E). However, it is important to note that NOTCH3 ICD was detectable upon exposure to Notch ligands, and NOTCH2 ICD was present even in unstimulated cultures; therefore, redundant signaling by these receptors *in vivo* might explain the results observed in the NOTCH1 KO cohort.

#### Satellite cell engraftment by iPax3-N1OE and iPax3-N1KO myogenic progenitors

Next, we asked whether iPax3-N1OE and iPax3-N1KO myogenic progenitors have the ability to seed the satellite cell pool. Immunofluorescence staining showed the presence of donor-derived RFP+ nuclei expressing PAX7 under the basal lamina in all cohorts (Figures S3F–S3I). Of note, the large majority of the satellite cell pool was donor-derived since irradiation prior to transplantation depleted the recipient's satellite cell pool (Heslop et al., 2000). We quantified the donor-derived satellite cell fraction by flow cytometry, identified as RFP+CD31<sup>−</sup>CD45<sup>−</sup>ITGA7+VCAM-1+. As shown in Figures 3A and 3B, the frequency of donor-derived satellite cells was significantly higher in iPax3-N1OE grafts, while no differences were observed between iPax3-N1KO and iPax3 groups, suggesting that NOTCH1 is not required for iPax3 myogenic progenitors to seed the satellite cell pool but activation of Notch1 signaling enhances this ability.

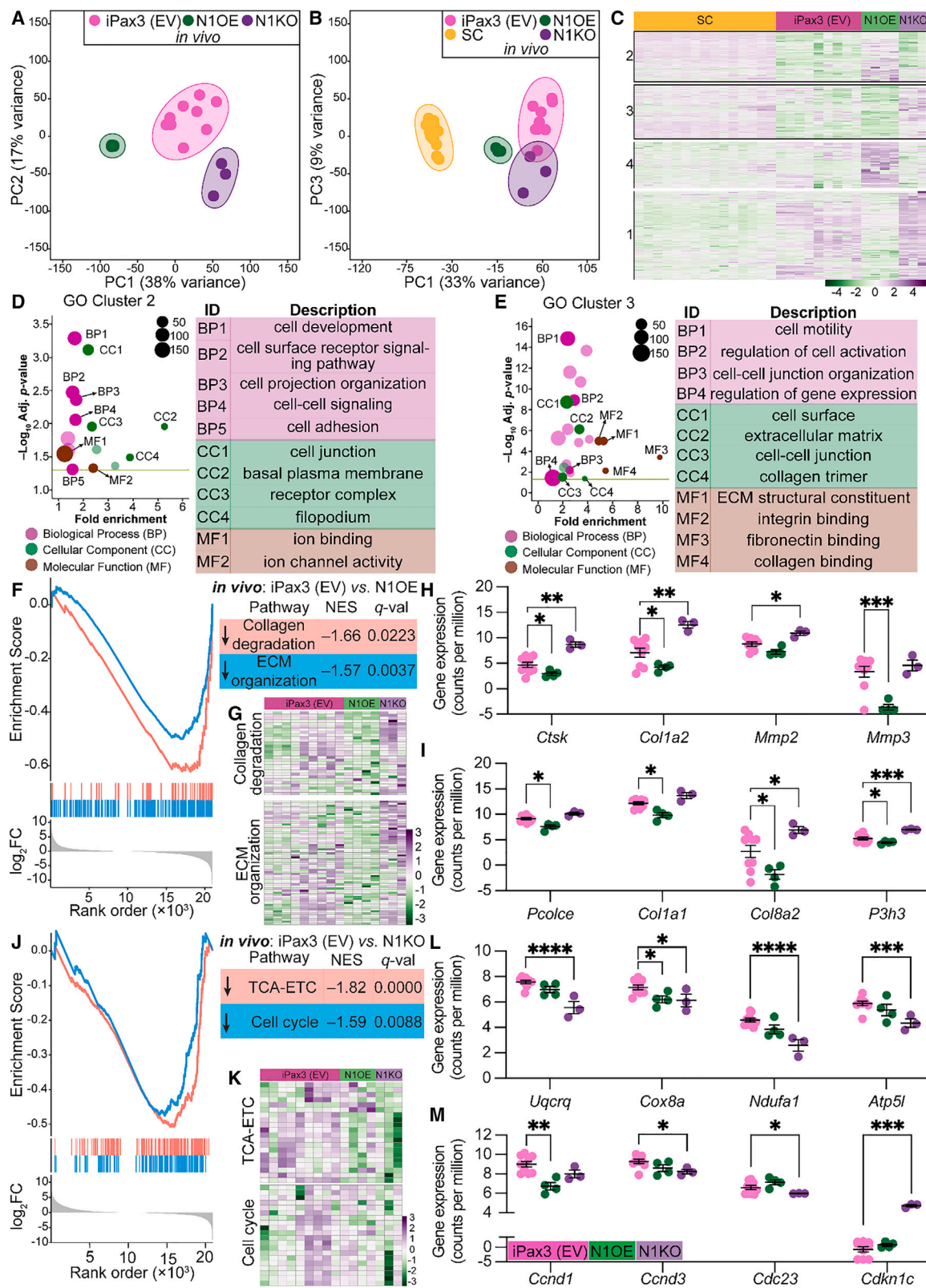
We confirmed NOTCH1 GoF and LoF in donor-derived satellite cells (Figure 3C), as shown by the upregulation and downregulation of Notch target genes in iPax3-N1OE and iPax3-N1KO re-isolated satellite cells, respectively (Figures 3C and 3D). We also looked at the expression of genes associated with satellite cell quiescence, such as *Pax3*, *Pax7*, *Myf5*, and *calcitonin receptor*, and found these to be present at higher levels in iPax3-N1OE satellite cells when compared to iPax3 controls (Figure S3J). These results are in agreement with previous studies demonstrating that adult satellite cell quiescence is linked to active Notch signaling (Bjornson et al., 2012; Fukada et al., 2011; Mourikis et al., 2012b) and NOTCH1 overexpression (NICD1) with upregulation of *Pax7* expression (Wen et al., 2012). On the other hand, iPax3-N1KO satellite cells showed comparable gene expression profile to iPax3 counterparts, except for *Pax3*, which was also elevated in this cohort, similar to iPax3-N1OE (Figure S3K). While less attention has been given to *Pax3* and Notch signaling, Conboy and Rando (2002) have shown that activation of Notch1 is associated with upregulation of *Pax3*. Since satellite cells from both NOTCH1 GoF and LoF cohorts exhibit high levels of *Pax3* compared to iPax3 controls, we anticipate that there is redundancy with other Notch receptors, as discussed previously for eMHC expression.

#### Regenerative capacity of iPax3-N1OE and iPax3-N1KO satellite cells

To assess the functionality of these engrafted satellite cells, we performed secondary transplantation assays by injecting 5,000 freshly isolated donor-derived RFP+CD31<sup>−</sup>CD45<sup>−</sup>ITGA7+VCAM-1+ satellite cells from each cohort into secondary recipients (as outlined in Figure 3E). Our results show comparable myofiber engraftment upon the secondary transplantation of satellite cells from all cohorts (Figures S4A–S4C). Fiber-type composition analysis of iPax3-N1OE and iPax3-N1KO secondary grafts showed comparable results to iPax3 controls, except for the

#### Figure 3. Satellite cell engraftment and function of iPax3-N1OE and iPax3-N1KO cells

(A and B) Representative FACS plots show the frequency of donor-derived ITGA7+VCAM-1+ satellite cells gated from the RFP+ and CD31<sup>−</sup>CD45<sup>−</sup> cell fractions in iPax3, iPax3-N1OE, and iPax3-N1KO muscle grafts (A). Graph shows the percentage of donor-derived RFP+CD31<sup>−</sup>CD45<sup>−</sup>ITGA7+VCAM-1+ in the total MNC fraction (B). Data are presented as mean ± SEM of two independent experiments ( $n = 11$  TAs for iPax3 controls,  $n = 4$  TAs for N1OE, and  $n = 4$  TAs for N1KO). \*\*\*\* $p < 0.0001$  by ANOVA with Dunnett's correction. (C and D) Bar graphs show gene expression for *Notch1* and downstream target genes: *Hes1*, *Hey1*, and *HeyL* in iPax3-N1OE (C) and iPax3-N1KO (D) re-isolated satellite cells along with iPax3 controls. Results are normalized to *Gapdh*. Data are shown as mean ± SEM of two independent experiments ( $n = 5–10$  TA muscles per sample). \* $p < 0.05$ , \*\* $p < 0.01$ , and \*\*\* $p < 0.001$  by unpaired Student's t test. (E) Outline of secondary transplantation studies and characterization. (F–I) Satellite cell engraftment. (F) Representative images show donor-derived satellite cell engraftment in iPax3, iPax3-N1OE, and iPax3-N1KO secondary recipients. Staining for RFP in red, PAX7 in green, and laminin (LAM) in white. DAPI in blue stained nuclei. Yellow circles indicate RFP+PAX7+ cells under the basal lamina, while arrows indicate the presence of RFP+PAX7+ cells in the interstitial space. Scale bar, 50  $\mu$ m. Graphs show quantification of the total number of RFP+PAX7+ cells per section (G) and the percentages of RFP+PAX7+ cells under the basal lamina (H) and at the interstitial (I) position. Data are presented as mean ± SEM of two independent experiments ( $n = 10$  TAs for iPax3 controls,  $n = 5$  TAs for N1OE, and  $n = 4$  TAs for N1KO). \*\* $p < 0.01$  and \*\*\*\* $p < 0.0001$  by ANOVA with Dunnett's correction.



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expression of MHC-I, which was lower in the iPax3-N1OE cohort (Figures S4D–S4E). These data suggest that, for most part, NOTCH1 GoF and LoF in donor-derived satellite cells do not affect myofiber type composition.

On the other hand, while assessment of satellite cell engraftment in these secondary recipients showed the presence of RFP+PAX7+ under the basal lamina in all cohorts (Figure 3F), this was found to be significantly lower in the iPax3-N1KO group (Figure 3H). This cohort displayed higher numbers of RFP+PAX7+ cells/section (Figure 3G), and the vast majority of these were found in the interstitial space (Figure 3I), indicating that Notch1 is required for proper seeding in the satellite cell niche. Of note, we observed no significant differences in proliferation between iPax3-N1KO and iPax3 controls, as indicated by Ki67 staining of RFP+ cells ( $2.6\% \pm 0.4\%$  and  $3.25\% \pm 0.5\%$ , respectively), suggesting that higher numbers of PAX7+RFP+ cells in iPax3-N1KO muscle grafts are not due to enhanced proliferation.

#### Molecular signature of donor-derived iPax3-N1OE and iPax3-N1KO satellite cells

To elucidate the molecular changes elicited by manipulating the Notch pathway in these myogenic progenitor transplants, we performed bulk RNA sequencing analysis of donor-derived satellite cells, as outlined in Figure 3E. Using linear dimensionality reduction of the top ~2,400 DEGs, principal component analysis (PCA) revealed clustering of iPax3 controls between iPax3-N1OE and iPax3-N1KO satellite cells (Figure 4A), with a separation along the PC1 axis between iPax3 control and Notch1-manipulated cells. For reference, we included primary satellite cells (here abbreviated as SCs) in the PCA analysis, using publicly available datasets (Figure 4B) (Bianchi et al., 2020;

Collins et al., 2019; Lilja et al., 2017; Shams et al., 2022; Verma et al., 2018; Yue et al., 2020). The differences between primary SCs and satellite cells derived from transplanted myogenic progenitors are evident in PC1, and Notch1 overexpression shifts these cells to the left, i.e., toward primary SC (Figure 4B).

Unsupervised hierarchical clustering analysis of the top 2,398 DEGs identified four clusters (Figure 4C). We focused on clusters 2 and 3 since they contained enriched genes that were upregulated in Notch GoF (cluster 2) and LoF (cluster 3) cohorts, as well as in SCs. GO analysis of cluster 2 showed significant enrichment of biological processes related to cell development, cell surface receptor signaling pathway, cell projection organization, cell-cell signaling, and cell adhesion (Figure 4D), whereas cluster 3 was enriched for biological processes related to cell motility, regulation of cell activation, cell-cell junction organization, and regulation of gene expression, among others (Figure 4E).

To get a better understanding of more transcriptionally global changes upon Notch1 modifications, we performed GSEA by combining the annotations from the Reactome and KEGG pathway databases. We found that iPax3-N1OE satellite cells exhibit significantly lower levels of genes related to cell-cell communication (Figure S4F), collagen degradation, and ECM organization signaling pathways compared to iPax3 (Figure 4F). Next, we retrieved the top most variable genes annotated to each of these signaling pathways among all the experimental groups to get a more granular understanding of the specific genes underlying these changes (Figures 4G and S4G). For the collagen degradation pathway, we found that expression levels of *cathepsin K* (*Ctsk*), *collagen type I alpha 2 chain* (*Col1a2*), and *matrix metalloproteinase 3* (*Mmp3*) were lower in the context of Notch1 GoF, while the LoF counterpart

#### Figure 4. Transcriptional signature of donor-derived iPax3-N1OE and iPax3-N1KO satellite cells

- (A) PCA plot of donor-derived satellite cells from iPax3 (EV) control (magenta), iPax3-N1OE (green), and iPax3-N1KO (purple) transplanted mice.
- (B) The same PCA with the inclusion of primary satellite cells (SCs, golden).
- (C) RNA sequencing hierarchical clustering of the top 2,398 DEGs shared between all samples. Clusters are indicated on the left margin.
- (D) Gene ontology analysis for cluster 2.
- (E) Gene ontology analysis for cluster 3.
- (F) GSE analysis with downregulated (down arrow) signaling pathways for iPax3 (EV) vs. N1OE.
- (G) Representative hierarchical clustering of most variable genes annotated for collagen degradation ( $n = 30$  genes) and ECM organization ( $n = 60$  genes) pathways.
- (H and I) Bar plots of gene expression levels for representative genes annotated to collagen degradation signaling (H) and ECM organization (I) pathways. Data are presented as mean  $\pm$  SEM ( $n = 9$  TAs for iPax3 in magenta,  $n = 4$  TAs for N1OE in green, and  $n = 3$  TAs for N1KO in purple).  $*p < 0.05$ ,  $**p < 0.01$ , and  $***p < 0.001$  by ANOVA with Dunnett's correction.
- (J) GSE analysis with downregulated (down arrow) signaling pathways for iPax3 vs. N1KO.
- (K) Representative hierarchical clustering of most variable genes annotated for TCA-ETC ( $n = 20$  genes) and cell cycle ( $n = 20$  genes) pathways.
- (L and M) Bar plots of gene expression levels for representative genes annotated to TCA-ETC (L) and cell cycle (M) pathways. iPax3 (magenta), N1OE (green), and N1KO (purple). Data are presented as mean  $\pm$  SEM ( $n = 9$  TAs for iPax3 in magenta,  $n = 4$  TAs for N1OE in green, and  $n = 3$  TAs for N1KO in purple).  $*p < 0.05$ ,  $**p < 0.01$ ,  $***p < 0.001$ , and  $****p < 0.0001$  by ANOVA with Dunnett's correction.



displayed the opposite phenotype with upregulation of *Ctsk*, *Col1a2*, and *Mmp2* (Figure 4H; Table S1 includes comparisons to SCs). In terms of ECM organization, we observed a similar trend with iPax3-N1OE satellite cells showing lower expression levels of *procollagen C-endopeptidase enhancer*, *collagen type I alpha 1 chain*, *collagen type VIII alpha 2 chain* (*Col8a2*), and *prolyl 3-hydroxylase 3* (*P3h3*), whereas iPax3-N1KO exhibited enhanced expression of *Col8a2* and *P3h3* (Figure 4I). Genes associated with cell-cell communication, such as *cadherin 11* (*Cdh11*), *cell adhesion molecule 1*, *Par-6 family cell polarity regulator gamma* and *Parvin beta*, were found downregulated in iPax3-N1OE compared to iPax3 controls, while iPax3-N1KO satellite cells showed increased expression levels of *Cdh11* (Figure S4H).

When assessing the differences between iPax3-N1KO and iPax3 control groups, we observed that iPax3-N1KO satellite cells displayed significant downregulation of tricarboxylic acid and electron transport chain (TCA-ETC) and cell cycle pathways (Figures 4J and 4K). The following genes belonging to the TCA-ETC pathway were found to be downregulated in the Notch1 LoF, *ubiquinol-cytochrome C reductase complex III subunit VII* (*Uqcqr*), *cytochrome c oxidase subunit 8A* (*Cox8a*), *NADH:ubiquinone oxidoreductase subunit A1* (*Ndufa1*), and *ATP synthase membrane subunit G* (*Atp5l*) relative to iPax3 controls. The Notch1 GoF showed comparable expression levels to iPax3 controls, except for *Cox8a*, which was also found downregulated (Figure 4L). For the cell cycle pathway, we observed downregulation of *cyclin D3* and *cell division cycle 23* and upregulation of *cyclin-dependent kinase inhibitor 1C* in iPax3-N1KO satellite cells, while iPax3-N1OE satellite cells showed downregulation of *cyclin D1* (Figure 4M).

## DISCUSSION

Here, we investigated the effect of NOTCH1 GoF and LoF on the engraftment and *in vivo* maturation switch of PSC-derived iPax3 myogenic cells. Since *in-vitro*-generated iPax3 myogenic progenitors display low expression levels of several components of the Notch signaling pathway (Incitti et al., 2019), we did not necessarily anticipate a phenotype in the LoF arm of primary recipients. Indeed, this was the case as our results showed no differences in myofiber engraftment between iPax3 and iPax3-N1KO cohorts, suggesting that NOTCH1 is not required for muscle regeneration in this system. Likewise, more Notch1 signaling did not affect the regenerative ability of iPax3 myogenic progenitors. While studies in aged mice have shown that Notch activation improved muscle regeneration (Conboy et al., 2003), our experimental system is quite distinct as we are not assessing regeneration in the context of aging.

On the other hand, Wen and colleagues (Wen et al., 2012) have reported that muscle stem cell-specific overexpression of NICD1 results in enhanced self-renewal of satellite cells with impaired muscle regeneration (Wen et al., 2012). This contrasts from our findings, which show comparable engraftment in both primary and secondary transplantation cohorts. This could be due to differences in levels and duration of Notch1 activation, in addition to distinct experimental systems (transplantation vs. host regeneration). As previously pointed out, results from Notch activation are highly context dependent (spatiotemporal and/or the cellular status) (Hurlbut et al., 2007; Schwanbeck et al., 2011).

Regarding myofiber type, we have previously reported that transplantation of PSC-derived iPax3/iPax7 myogenic progenitors into irradiated CTX-injured TA muscles produces mostly myofiber grafts expressing embryonic, type I and type IIA MHC isoforms (Incitti et al., 2020). While the same trend was observed here for iPax3 controls, our data show that iPax3-N1OE primary muscle grafts display more mature fiber types, as evidenced by the blunt decrease in the expression levels of eMHC and increase in type IIX fibers. Unexpectedly, although to a lesser extent, muscles engrafted with iPax3-N1KO myogenic progenitors displayed a phenotype similar to the iPax3-N1OE cohort in regard to eMHC and IIX fibers, which is probably due to signaling redundancy by other Notch receptors. For instance, Fujimaki and colleagues have reported that Notch1 and Notch2 coordinately maintain the satellite cell pool (Fujimaki et al., 2018). While NOTCH3 is not expressed in iPax3 and iPax3-N1KO myogenic progenitors, our data show that both cell populations express comparable protein expression levels of NOTCH2 and that NOTCH3 becomes upregulated upon exposure to Notch ligands JAG1 and Dll4; thus, it is plausible that Notch2 and Notch3 signaling may account for the observed phenotype *in vivo*. The muscle fiber has been reported as a source of DLL4 (Eliazar et al., 2022). Another possibility is the involvement of the non-canonical Notch signaling pathway (Traustadóttir et al., 2016). Besides these changes, iPax3-N1KO grafts showed a significant reduction in type I and IIA MHC isoforms, suggesting a potential role for Notch1 signaling in the development/maintenance of oxidative fibers. Of note, in secondary transplantation assays, all cohorts (Notch1 GoF and LoF as well as iPax3 controls) exhibit comparable blunted expression of eMHC and increased type IIX fibers (Figure S4E), confirming our previous findings with iPax3 myogenic progenitors (Incitti et al., 2020) and reinforcing the premise that our PSC-derived myogenic progenitors remodel their molecular signature upon exposure to the adult muscle environment (Abreu et al., 2023; Incitti et al., 2019), which correlates with upregulation of Notch signaling. Of interest, Mourikis and



colleagues have reported that NICD-expressing cells of embryonic origin undergo maturation as they adopt features of fetal stage myogenic cells (Mourikis et al., 2012a).

Satellite cell engraftment is essential for long-term graft maintenance. Our data show that whereas disruption of Notch1 signaling did not affect the ability of *in-vitro*-generated iPax3 myogenic progenitors to seed the satellite cell compartment in primary transplantations, NICD1 overexpression led to an increased frequency of donor-derived satellite cells in primary recipients (about 3-fold, quantified by flow cytometry). In secondary grafts, NOTCH1 overexpression was still accompanied by a higher frequency of satellite cells, but this was less evident (quantification based on the number of PAX7+ under the basal lamina). In any case, the most remarkable phenotype was observed in iPax3-N1KO secondary recipients, as transplantation of satellite cells lacking NOTCH1 resulted in disturbed homing of PAX7+ cells to the satellite cell position. The vast majority of RFP+PAX7+ cells were found in the interstitial space of these secondary grafts (Figure 3F). Of note, Brohl and colleagues reported that double mutation of the major Notch signaling transcriptional mediator RBP-J and MYOD (double *Rbpj*/*MyoD* knockout) in Pax3 cells leads to disrupted homing of emerging satellite due to impaired cell adhesion and basal lamina assembly (Brohl et al., 2012). Our transcriptomic data of transplanted satellite cells showed significant enrichment of biological process related to cell projection organization, cell-cell signaling, cell adhesion, cell junction, filopodium, cell motility, and ECM. More precisely, using KEGG and Reactome pathways, we identified upregulation of genes associated with collagen degradation and ECM organization in iPax3-N1KO satellite cells, such as *Ctsk*, a lysosomal cysteine proteinase known for its potent collagenase activity (Zhang et al., 2024), and collagen alpha chains (*Col1a* and *Col8a2*), while these, among other genes, were found downregulated in the presence of Notch1 overexpression, suggesting that Notch1 regulation is important for ECM remodeling. *Mmp2*, zinc-dependent enzymes capable of cleaving components of the ECM (Cabral-Pacheco et al., 2020), were also found to be elevated in the absence of Notch1 signaling. We also observed changes in genes associated with cell adhesion, cell spreading, establishment/maintenance of cell polarity, and cell migration.

Overall, our RNA sequencing data support the premise that NOTCH1 ablation results in dysregulated ECM in iPax3 PSC-derived satellite cells. ECM is known to be important for satellite cell homing (Brohl et al., 2012; Rayagiri et al., 2018). The fact that the interstitial localization defect was observed only upon the transplantation of re-isolated donor-derived iPax3-N1KO satellite cells, not with the initial transplantation of *in-vitro*-generated PSC-derived iPax3-N1KO myogenic progenitors, further corroborates

that these cells acquire features of *bona fide* satellite cells (i.e., maturation) only after a period of exposure to the adult environment. Of note, using patient induced PSC-derived myogenic progenitors, Ortiz-Vitali and colleagues have recently demonstrated that mutations in POGLUT1 are associated with defective Notch signaling and mislocalization of PAX7+ cells (Ortiz-Vitali et al., 2023).

Taken together, these Notch1 GoF and LoF studies provide comprehensive insights into the role of Notch signaling in the engraftment and *in vivo* maturation of PSC-derived myogenic progenitors, highlighting the notion that engrafted PSC-derived myogenic cells undergo *in vivo* maturation upon exposure to Notch ligands within the muscle environment. This *in vivo* maturation occurs both at the level of the donor-derived putative satellite cells and at the level of the myofibers.

## METHODS

### Cell culture, differentiation, and Notch engineering

Detailed description of maintenance of iPax3 mouse embryonic stem (ES) cells is provided in supplemental methods.

For *in vivo* tracking, iPax3 ES cells were labeled with lentiviral vector encoding the fusion protein histone 2B-red fluorescent protein (H2B-RFP; LV-RFP plasmid, Addgene #26001), as previously described (Baik et al., 2023). See supplemental methods for more details.

For GoF studies, the murine *NICD1* (Plasmid #201983, Addgene) open reading frame was amplified by PCR and the sequence cloned into a pRRL-T2AGFP vector, and confirmed by DNA sequencing. The lentiviral vectors pRRL-NICD1-T2AGFP and pRRL-T2AGFP (control, empty vector) were co-transfected with pVSV-G and Δ8.9 packaging plasmids into HEK293T cells using the Lipofectamine LTX-Plus reagent (Invitrogen). Lentiviral-containing supernatants were collected 48 h after transfection, filtered, and used for transduction of myogenic progenitors at passage 0 by spin infection (90 min at 2,500 rpm at 30°C). After 2 days, GFP+ cells were purified using a FACSaria II, replated, and expanded in myogenic medium until passage 3, as described previously (Azzag et al., 2020).

For LoF studies, a CRISPR-Cas9 construct was generated by cloning single-guide RNA (sgRNA) (5'-AAACCTTG GCTGGGAGCATCTCAAC-3') targeting exon 2 of the Notch1 receptor (Mandasari et al., 2016) using the BbsI restriction enzyme sites into pSpCas9(BB)-2A-GFP (PX458) vector (Plasmid #48138, Addgene) (Ran et al., 2013). The sgRNA/Cas9 plasmids were delivered to iPax3 ES cells by nucleofection (Lonza) according to the manufacturer's protocol. Clones from KO cell cultures were obtained by





single-cell sorting based on GFP expression and expanded. We obtained approximately 10–20 Notch1 KO single cell clones. Deletion in selected clones was confirmed by DNA sequencing (data not shown).

### Mice and primary cell transplantation

All animal experiments were carried out according to protocols approved by the University of Minnesota Institutional Care and Use Committee. Hindlimbs of 6- to 12-week-old mice NOD-scid IL2R<sup>gnull</sup> (NSG) mice (Jackson laboratories) were irradiated with a 12-Gy single dose, and 24 h later, both TA muscles were injected with 15  $\mu$ L of 10  $\mu$ M CTX (Latoxan). One day after CTX injury, myogenic progenitors were transplanted into the left TA ( $3 \times 10^5$  cells/15  $\mu$ L), while the contralateral TA received PBS (control). At 4 weeks post transplantation, TAs were collected for the assessment of myofiber and satellite engraftment or for donor-derived satellite cell isolation, as detailed in the following text.

### Donor-derived satellite cell isolation and transplantation

Donor-derived satellite cell isolation was based on the expression of RFP (donor) and the following surface marker profile, CD31<sup>−</sup>CD45<sup>−</sup>ITGA7<sup>+</sup>Vcam<sup>+</sup> (Chan et al., 2018; Incitti et al., 2019). See [supplemental methods](#) for more details.

### Western blotting

Proteins were extracted using RIPA buffer (150 mM NaCl, 50 mM Tris-HCl, pH 7.5, 1 mM EDTA, 1% Triton, 1% sodium deoxycholate, and 0.1% SDS) supplemented with 1% protease inhibitors (Complete, Millipore). Lysates were centrifuged at 15,000 rpm for 5 min at 4°C. Total protein concentration was measured using by Bradford assay (Sigma-Aldrich). See [supplemental methods](#) for more details.

### Immunofluorescence staining

Immunofluorescence staining was performed on TA muscle cryosections and cultured cells. See [supplemental methods](#) for more details.

### Quantitative analysis of myotube differentiation

Quantification of several parameters was performed by immunofluorescence imaging of differentiated myotubes stained for pan-MHC and nuclei (DAPI). Cells were imaged using Zeiss inverted microscope, and analysis was performed from random fields of the well. The fusion index was calculated as the percentage of nuclei within myotubes relative to the total number of nuclei per microscopic field. Myotube width was measured by determining the maximum width ( $\mu$ m) of each myotube, for all myotubes

in each microscope field. Myotube area ( $\mu$ m<sup>2</sup>) was calculated by determining the proportion of the field occupied by positive pan-MHC staining, relative to the total field area.

### RNA isolation, quantitative reverse-transcription PCR, and RNA sequencing

RNA isolation was performed in two different ways. For samples with high input (*in vitro* cells), RNA was extracted using Trizol (Invitrogen) and RNA extraction kit (Invitrogen) following manufacturer's instructions. RNAs were retrotranscribed using SuperScript VILO cDNA Synthesis Kit (Invitrogen) following manufacturer's instructions. For samples with low input of RNA (*in vivo* re-isolated donor-derived satellite cells), RNA was extracted using Buffer RLT (QIAGEN) and nucleic acid beads. The cDNA was synthesized using the NEBNext Single Cell/Low Input cDNA Synthesis & Amplification kit (New England Biolabs).

Gene expression analyses were performed using 10 ng of cDNA for each reaction. The quantitative reverse-transcription PCR analysis was performed using Premix Ex Taq Probe qPCR Master Mix (Takara) and TaqMan probes from Applied Biosystems (*Notch1*, Mm01326741\_g1; *Notch3*, Mm01345646\_m1; *Hes1*, Mm01342805\_m1; *Hey1*, Mm00468865\_m1; *HeyL*, Mm00516558\_m1; and *Gapdh*, Mm05724508\_g1.) Gene expression levels were normalized by *Gapdh* levels.

For the RNA sequencing analyses, the libraries were generated with the Nextera XT DNA Library Preparation Kit (Illumina). These samples were processed at the University of Minnesota Genomics Center (UMGC). The libraries were then sequenced on a paired-end run on the Nova-Seq (Illumina).

### Satellite cell RNA sequencing datasets

We reviewed research articles from 2015 to 2024 that utilized freshly isolated and directly FACS-sorted *bone fide* murine satellite cells as defined by either expression of Pax7 or well-established satellite cell markers. Of these, we selected bulk RNA sequencing datasets that used paired-end libraries using the Illumina platform and downloaded the raw FASTQ files from the Gene Expression Omnibus (GEO) database using the Sequence Read Archive toolkit. This initial review resulted in a total of 21 biological samples from GSE205015 (Shams et al., 2022), GSE113631 (Yue et al., 2020), GSE123723 (Bianchi et al., 2020), GSE126872 (Collins et al., 2019), GSE108739 (Verma et al., 2018), and GSE89977 (Lilja et al., 2017). We then removed any biological samples with very low expression ( $\leq 10$  million reads) and/or high mitochondrial gene expression contamination, which resulted in a final set of 15 biological samples.



### RNA sequencing processing and analysis

We used our well-established pipeline, as previously published (Abreu et al., 2023). Briefly, we conducted data alignment using the CHURP pipeline (Baller et al., 2019) at the UMGC. 2 × 150 bp FASTQ paired-end reads for a total of 23 biological samples (~20 million protein-coding mapped reads average per sample) were trimmed using Trimmomatic (v.0.33) enabled with the optional “-q” option; 3 bp sliding-window trimming from 3′ end requiring minimum Q30, and additional headcrop of 3 bp for samples processed with Takara Bio during library preparation. Quality control on raw sequence data for each sample was performed with FastQC (RIN ≥ 9.7). Read mapping was performed via HISAT2 (Kim et al., 2019) (v.2.1.0) using the mouse genome (GRCh39.106) as reference. Gene quantification was done via Feature Counts for raw read counts. Batch correction to adjust for differences in the library preparation (TruSeq Stranded vs. SMARTer Stranded) and batch effects were performed using the CombatSeq (Zhang et al., 2020) package (v.3.46.0) on R. Differentially expressed genes were identified using the EdgeR (negative binomial, R programming) feature in CLCGWB (QIAGEN, Redwood City, CA) using batch-corrected raw read counts. We filtered the generated list based on a minimum 2x absolute fold change and false discovery rate (FDR)-corrected  $p < 0.05$ . Gene set enrichment (GSE) analysis was performed using the clusterProfiler (Wu et al., 2021; Yu et al., 2012) package (v.4.6.2) on R using both KEGG (Kanehisa and Goto, 2000) and Reactome (Wu and Haw, 2017) pathway databases for annotations. Gene ontology (GO) analysis was carried out using the DAVID Knowledgebase (Huang et al., 2009; Sherman et al., 2022) (v. Apr, 2024) Functional Annotation Tool, which reduces the dependency between terms. Pathways and GO terms with an FDR  $q < 0.05$  were considered to be significantly expressed.

### Quantification and statistical analysis

Statistical differences between two independent samples were analyzed using the unpaired two-tailed Student's  $t$  test. Statistical differences between three groups were assessed using one-way or two-way ANOVA followed by post hoc correction for multiple comparisons using Tukey's or Dunnett's correction.  $p$  values lower than 0.05 were considered significant. GraphPad Prism v9 software (GraphPad Software, LLC) was used for statistical analyses.

### RESOURCE AVAILABILITY

#### Lead contact

Further information and requests for resources and reagents should be directed to and will be provided by the lead contact, Rita Perlingeiro (perli032@umn.edu).

### Materials availability

Plasmids and cell lines generated in this study will be provided upon request to Rita C.R. Perlingeiro (perli032@umn.edu).

### Data and code availability

Flow cytometry, confocal, qPCR, and inverted microscopy raw data files reported in this paper will be shared by the lead contact upon request. Any additional information required to reanalyze the data reported in this paper is available by the lead contact upon request. RNA-sequencing data generated in this study has been deposited at GEO and are publicly available as of the date of publication (accession number GSE271221). This paper does not report original code.

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### AUTHOR CONTRIBUTIONS

A.M.S.Y. designed and performed the experiments, analyzed the data, and wrote the manuscript. B.I.G. analyzed the data and wrote the manuscript. H.K. designed and performed the experiments and contributed to data analysis. J.E.A. and K.A. analyzed the data. D.B., P.A., and A.A. performed the experiments. R.C.R.P. contributed with experimental design, interpretation of the data, and wrote the manuscript.

### DECLARATION OF INTERESTS

R.C.R.P. is the cofounder and holds equity in Myogenica.

### SUPPLEMENTAL INFORMATION

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